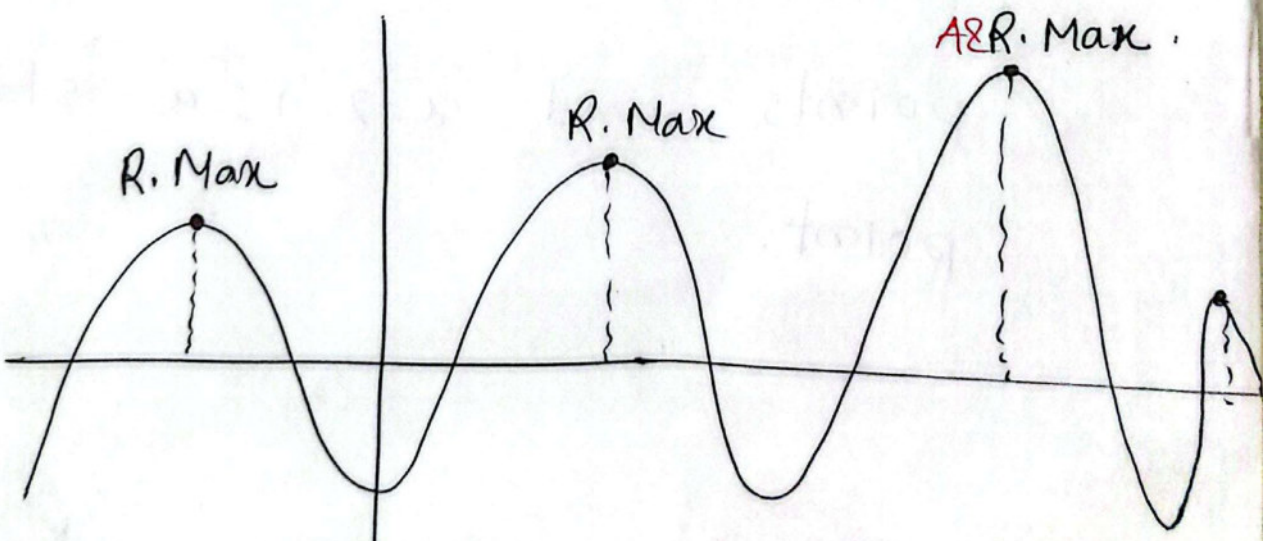
* Relative Maximum or Relative Min. using First derivative test:

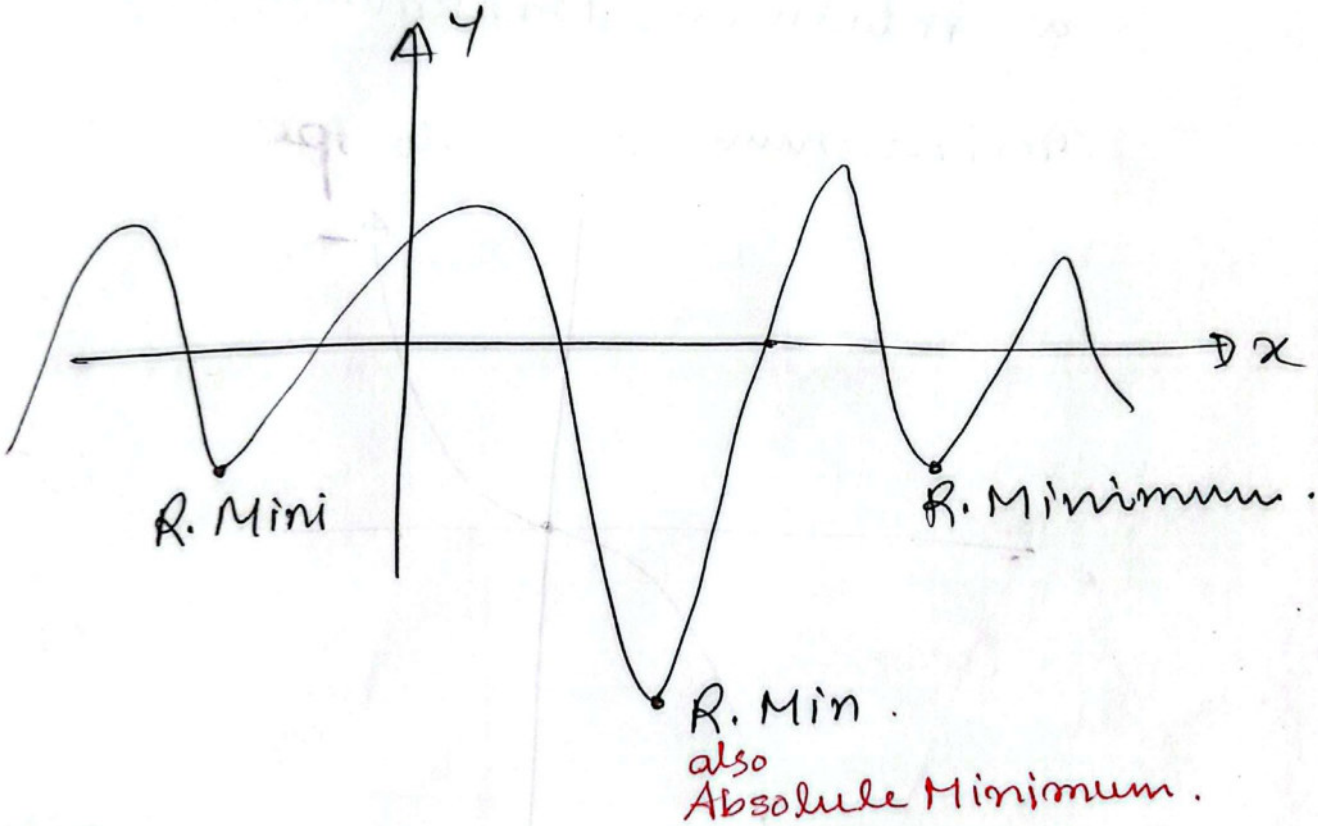
Let us ~~if~~ suppose that f is continuous at a critical point x_0.

a) If $f'(x) > 0$ on an open interval extending left from x_0 and $f'(x) < 0$ on an open interval extending right from x_0, then f has a relative Maximum at x_0.



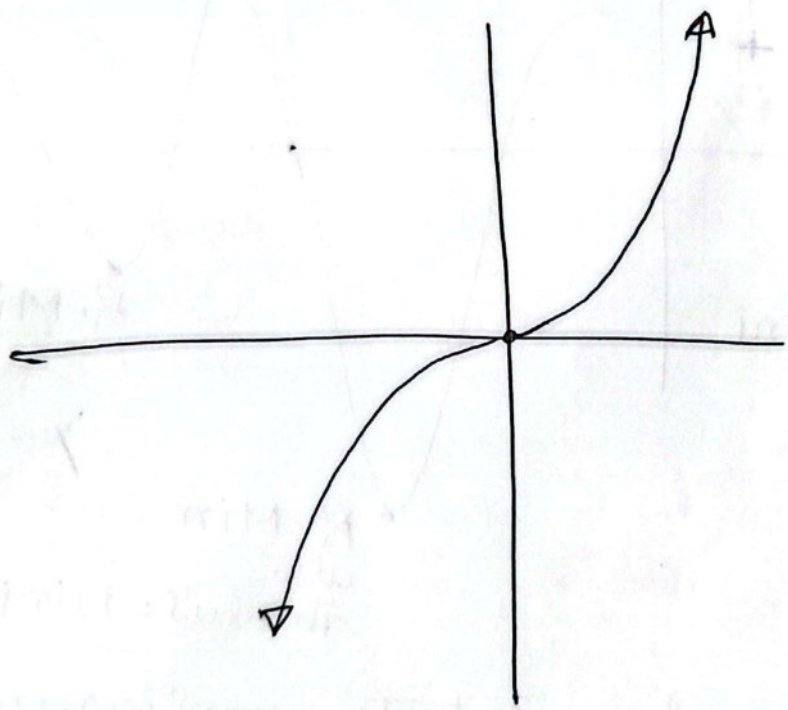
A relative Maximum point is a point where the function changes direction from increasing to decreasing.

b) If $f'(x) < 0$ on an open interval extending left from x_0 and $f'(x) > 0$ on an open interval extending right from x_0, then f has a relative Minimum at x_0.



That is, A relative minimum point is a point where the function changes direction from decreasing to increasing.

(c) If $f'(x)$ has same sign on an open interval extending left from x_0 as it does on an open interval extending right from x_0, then f does not have a relative Maximum nor relative minimum at x_0.



<u>Example</u>: the function $f(x) = 3x^{5/3} - 15x^{2/3}$ has a relative maximum at $x=0$ and a relative minimum at $x=2$. confirm this using first derivative test.

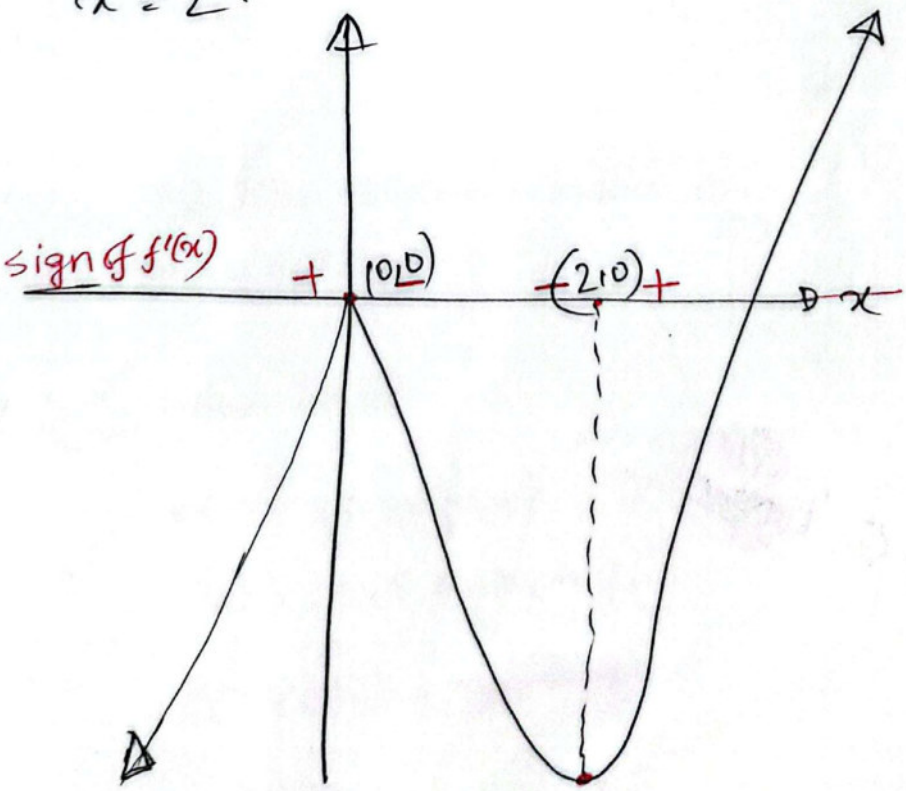
Solution

$$f(x) = 3x^{5/3} - 15x^{2/3}.$$

$$f'(x) = \frac{5(x-2)}{x^{1/3}}$$

The sign of f' changes from '+' to '−' at $x=0$, so ~~tha~~ there is a relative maximum at $x=0$.

The sign of f' changes from '−' to '+' at $x=2$, so there is a relative minimum point at $x=2$.



If : $f'(x_0)=0,\ f''(x_0)=0 \ldots\ldots f^{(n)}(x_0)=0,\ n=2,3,\ldots,k-1$

$f^{k}(x_0)\neq 0$ where $k\geq 2$

If k is even and $f^{k}(x_0)>0 \Rightarrow$ Local Mi

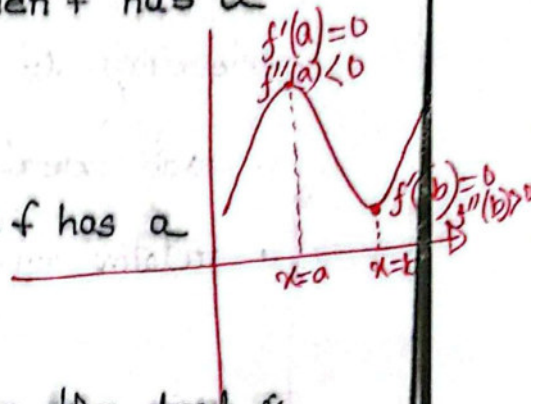
" " " " " $f^{k}(x_0)<0 \Rightarrow$ L. Maxi

" k is odd then no extremum

Second Derivative Test:— Let us suppose that f is twice differentiable at the point x_0.

$f(x)=x^4$

a) If $f'(x_0)=0$ and $f''(x_0)>0$, then f has a relative minimum at x_0.



b) If $f'(x_0)=0$ and $f''(x_0)<0$, then f has a relative maximum at x_0.

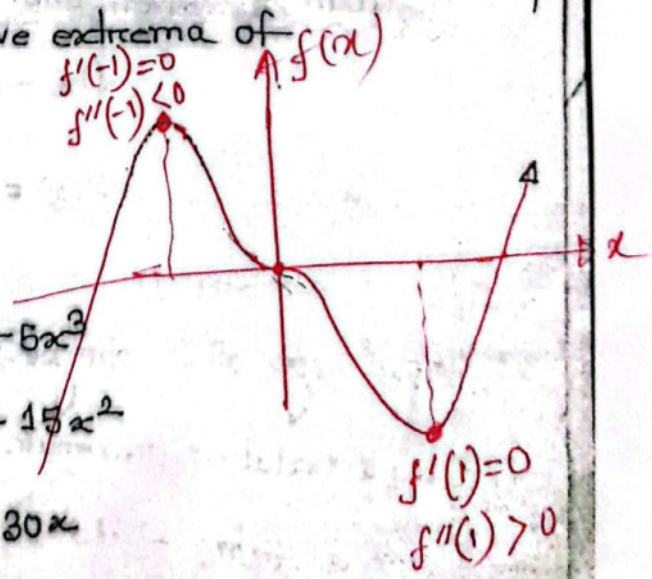
c) If $f'(x_0)=0$ and $f''(x_0)=0$ then the test is inconclusive, i.e. f may have a relative maximum, a relative minimum, or neither at x_0.

$f'(x)\neq 0,\ f''(x)=0,\ f(x)=x+2$

$f'(x)=0,\ f''(x)=0\ \ f(x)=c$

Example:— Find the relative extrema of

$$f(x)=3x^5-5x^3.$$



Solution:— Given that

$$f(x)=3x^5-5x^3$$

$$f'(x)=15x^4-15x^2$$

$$f''(x)=60x^3-30x$$

$$=30x(x^2-1)$$

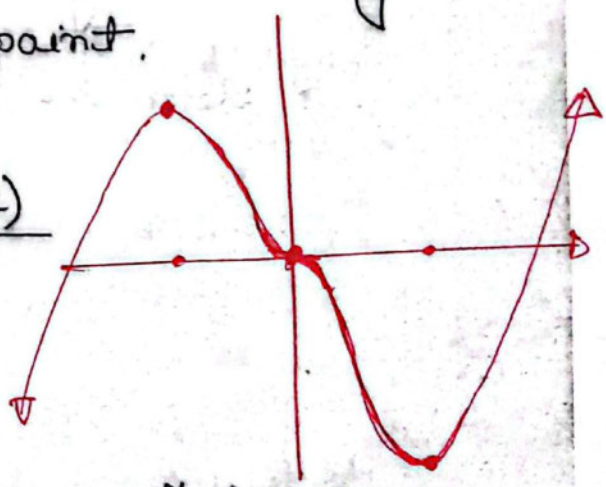
Solving $f'(x) = 0$.

$\Rightarrow x = 0, -1, 1 \rightarrow$ Stationary Points

particular points.	$f''(x) = 30x(2x^2-1)$	sign of $f''(x)$	Second derivative Test
$x = -1$	-30	$-$	f has a relative maximum
$x = 0$	0	0	Inconclusive
$x = 1$	30	$+$	f has relative minimum

The test is inconclusive at $x=0$, so we will try the first derivative test at that point.

Interval	$15x^2(x+1)(x-1)$	$f'(x)$
$-1 < x < 0$	$(+)(+)(-)$	$-$
$0 < x < 1$	$(+)(+)(-)$	$-$

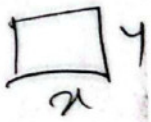


Since there is no sign change in f' at $x=0$, there is neither a relative maximum nor a relative minimum at that point.

General Guidelines for Sketching a Graph of a Function :-

1. Domain
2. Intercepts
3. Symmetry Test
4. Asymptotes
5. Intervals of increase and decrease
6. Local maximum and minimum
7. Concavity and Points of inflection
8. Sketch the curve.

Find $x+y$.



$x+y=5$

Area

$x=1, y=4$

$xy=4$

$x=2, y=3$

$xy=6$

$$\frac{d^r w}{dx^r} = k w^r$$